

Contextualization

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. A set $\mathcal{S}$ of such matrices with a given arbitrary p.d.f. $\mathcal{P}$ is called an ensemble. While studying eigenvectors may be of interest in some cases, eigenvalue statistics have been considerably more explored for these matrices. Hence, in addition to the p.d.f.s, we are also interested in the eigenvalue j.p.d.f. $\mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n)$, and the subsequent univariate eigenvalue density $\mathbf{p}(\lambda_1; n)$.

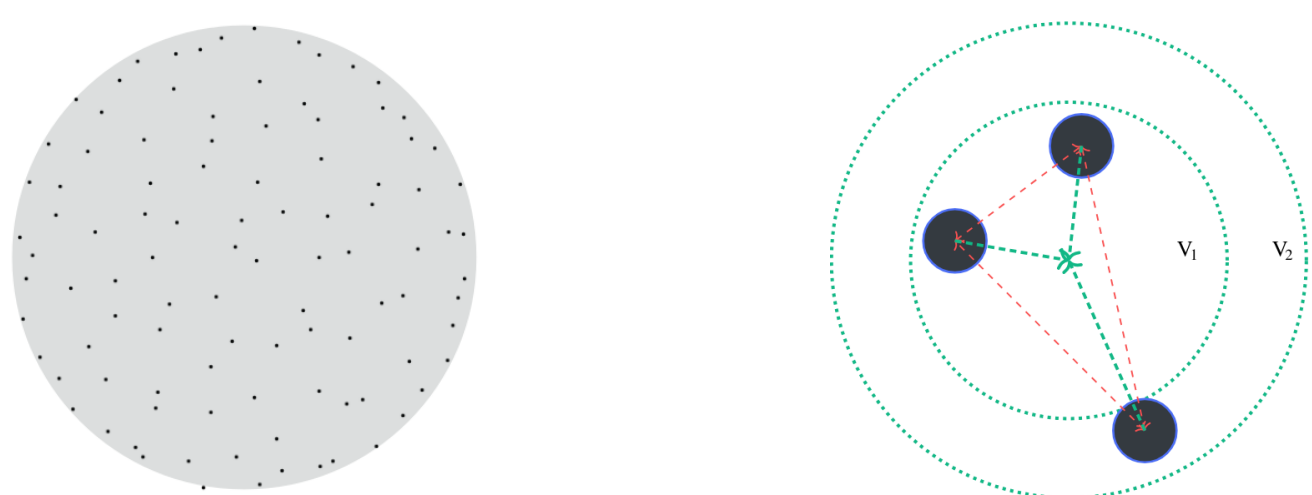
Here, *invariant matrix ensembles* refers to ensembles whose distribution can be expressed as a function of its eigenvalues. They are invariant by some appropriate group action. We are mainly interested in the subset called *Orthogonal Polynomial Ensembles* when $\beta = 2$ - that is, sets of matrices with j.p.d.f.

$$\mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{\mathcal{Z}_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta$$

for some continuous, non-negative, real-valued weight function $w(\lambda)$ and

$$\Delta_n(\lambda) := \det \left[\lambda_i^{j-1} \right]_{i,j=1}^n = \prod_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)$$

being the Vandermonde determinant. Such ensembles have a natural interpretation as a *Coulomb Gas* - an interacting gas of indistinguishable particles under an external potential.



The weight corresponds to an external potential and the determinant to the appropriate Coulombian interaction kernel. One can actually show that for these ensembles $\mathbf{p}(\lambda; \beta)$ is given by the formula

$$\begin{aligned} \mathbf{p}(\lambda; 2) &= K_n(\lambda, \lambda) \\ &= \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{n_k} q_k^{(2)}(x) q_k^{(2)}(y), \end{aligned}$$

where $\{q_j^{(\beta)}\}_{j=0}^\infty$ is a family of monic polynomials orthogonal in respect to $w(x)$, such that each $q_j^{(\beta)}(\lambda)$ is of degree j . Then, there exist finite positive constants, called norming constants, $\{n_j\}_{j=0}^\infty$ s.t.

$$\langle q_j, q_k \rangle^{(2)} = n_j \chi_{j=k}.$$

References

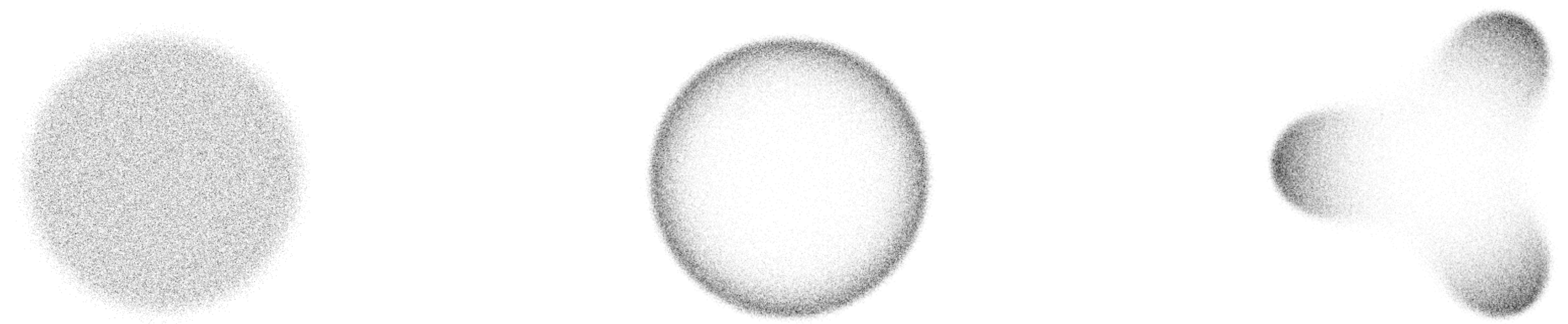
- [1] Sung-Soo Byun, Nam-Gyu Kang, and Seong-Mi Seo. Partition functions of determinantal and pfaffian Coulomb gases with radially symmetric potentials. *Communications in Mathematical Physics*, 401(2):1627–1663, March 2023.
- [2] Thomas Leblé and Sylvia Serfaty. Large deviation principle for empirical fields of log and Riesz gases. *Invent. Math.*, 210(3):645–757, December 2017.

Models of Interest

We will focus on a model related to Coulomb Gases in the complex plane, one could write explicitly its partition function, that is, the normalization for the distribution

$$\mathcal{Z}_{N,\beta}^Q := \int_{\mathbb{C}^N} \prod_{j>k=1}^N |z_j - z_k|^\beta \prod_{j=1}^N e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j).$$

where $dA(z) := d^2z/\pi$ is the area measure. This is the expression, if $\beta = 2$, for the eigenvalue distribution of the so-called **random normal matrices**. For $Q(z) = |z|^2$, for example, this would be the **complex Ginibre ensemble**, the simplest of the *Gaussian Ensembles*, arguably the most notorious class of ensembles.



We consider some potential radially symmetric $Q(z) : \mathbb{C} \rightarrow \mathbb{R}$ which satisfies some nice growth condition with $q(|z|) = Q(z)$. In a recent work [1] large- N asymptotic expansions were derived up to the $O(1)$ -terms for $\log(\mathcal{Z}_{N,\beta}^Q)$ given by

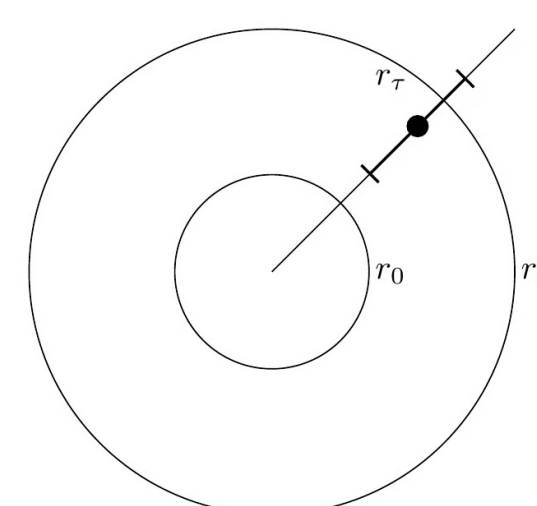
$$\log(\mathcal{Z}_{N,\beta=2}^Q) = -I^Q N^2 + \frac{1}{2} N \log N + E^Q N + G \log N + F^Q + o(1),$$

as $N \rightarrow \infty$. For a large class of potentials Q , the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \log N$, as well as the entropy term E^Q , were known from recent work of Leblé and Serfaty [2], also for rather large classes of Q . The remaining terms in this expansion are new. The term G appears to depend only on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the previous literature. Using some algebraic manipulation on the partition function, one can write the free energy for those models as

$$\log \mathcal{Z}_N^Q = \log N! + \sum_{j=0}^{N-1} \log h_j, \quad h_j := \int_{\mathbb{C}} |z|^{2j} e^{-NQ(z)} dA(z).$$

If we're interested in the coefficients for the asymptotic expansion for the free energy, it is enough to understand the asymptotic of the summation of the norming constants.

Annulus



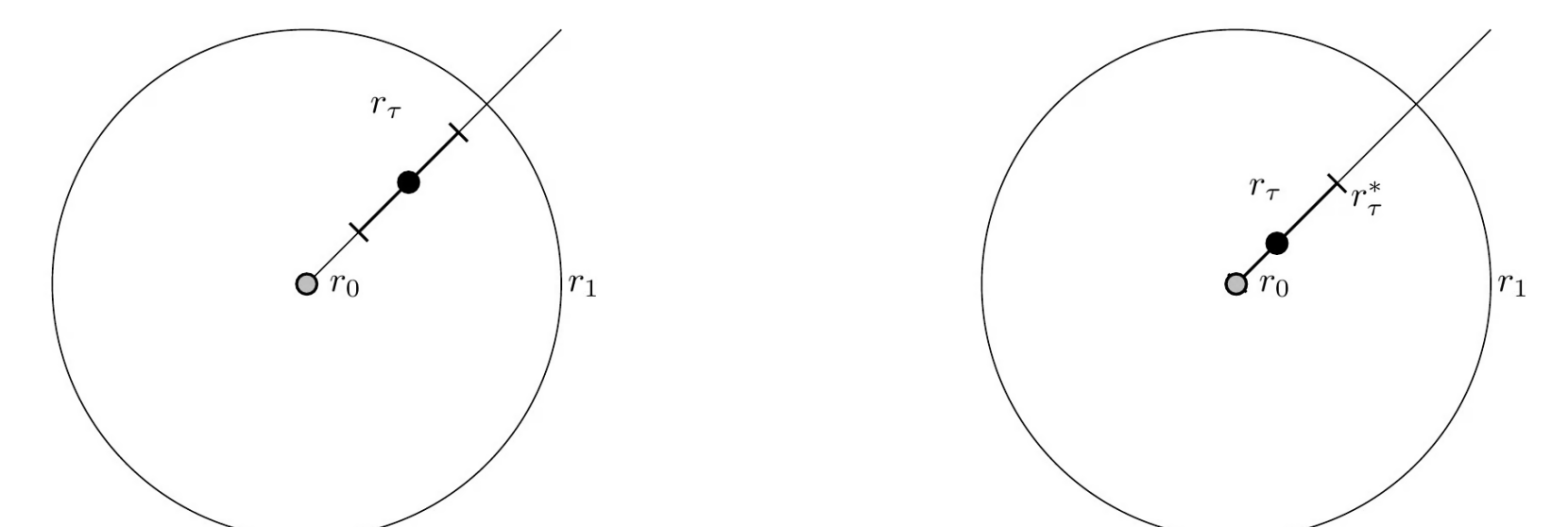
The first step in the asymptotic analysis is to change to radial variables $|z| = r$ and set $V_\tau(r) = q(r) - 2\tau \log(r)$ with $\tau(j) = j/N$. Then, we divide the integral using $\delta_N = \log(N)/\sqrt{N}$ for each h_j , as follows

$$\int_{|r-r_\tau|>\delta} e^{-NV_\tau(r)} 2r dr + \int_{|r-r_\tau|<\delta} e^{-NV_\tau(r)} 2r dr.$$

This is typical from the Laplace method - the first term gives the major contribution, the second corresponds to an error term, controlled in the asymptotic analysis. Then, to estimate the summation on $\log(h_j)$, we use the trapezoidal quadrature

$$\begin{aligned} \int_m^n f(x) dx &= \sum_{j=m}^n f(j) - \frac{f(m) + f(n)}{2} \\ &\quad - \sum_{k=1}^l \frac{B_{2k}}{(2k)!} (f^{(2k-1)}(n) - f^{(2k-1)}(m)) + R_l \end{aligned}$$

Disc



An important observation made is that the asymptotic behavior of h_j depends on whether the degree j is sufficiently small or not. The case where j is big enough, the method is virtually the same as before - we are dealing with almost the same local contributions. The interesting part comes when $j > N^\epsilon$ for $0 < \epsilon < 1/5$. Set $r_\tau^* := r_\tau \cdot \log(N)$ and divide the integral again

$$\int_{r < r_\tau^*} e^{-NV_\tau(r)} 2r dr + \int_{r > r_\tau^*} e^{-NV_\tau(r)} 2r dr.$$

The first term gives the main contribution to the asymptotic and the second can be controlled. Near the center we should expect some other regimen, which is observed in the results. As mentioned, up to the N -ordered term, all is the same. The difference shows from the $\mathcal{O}(\log(N))$ and $\mathcal{O}(1)$ terms - related to the genus of the support. This difference appears to have not been considered before.

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